Natural Gas Market Analysis — 2026

Python portfolio project | U.S. EIA weekly East Region storage and daily Henry Hub spot prices

Project overview

This project retrieves EIA time-series data, measures weekly natural gas storage changes, summarizes seasonality, aligns daily Henry Hub spot prices to Friday-ending weekly averages, and explores the statistical association between East Region storage changes and prices. It also examines the unusual January 2026 price spike.

Methods and datasets

Storage: EIA weekly working gas, East Region (duoarea R31; process SWO). Price: EIA daily Henry Hub spot series RNGWHHD. Storage changes are first differences in Bcf; daily prices are averaged into Friday-ending weeks before matching dates. The original exploratory script covered 2024–2026 storage and 2026 prices. The cleaned script requests observations through October 11, 2026.

Results from the completed Spyder session

Measure	Observed result
Matched weekly observations	39
Pearson correlation (storage change vs price)	–0.608
Correlation excluding Jan 23 and Jan 30	–0.539 (37 weeks)
Simple regression slope	–0.0450 $/MMBtu per Bcf
Simple regression intercept	3.7065 $/MMBtu
Regression R^2	0.3697
Largest observed weekly withdrawal	–75 Bcf (Jan 30, 2026)

Interpreting the results

The negative correlation indicates that, within these matched weeks, greater storage withdrawals tended to coincide with higher Henry Hub prices. $R^2 \approx 0.37$ means the one-variable linear fit describes approximately 37% of observed price variation in this sample. Removing the two highlighted weeks weakens the correlation, suggesting unusual winter conditions influence the fit. These relationships do not establish causation or predictive reliability.

January 23–27 anomaly: what can we conclude?

The exploratory chart highlights January 23–27, 2026 and labels an exceptional Henry Hub price near $30.72/MMBtu. The January 30 weekly storage change was −75 Bcf, and the corresponding weekly mean Henry Hub price in the Spyder output was approximately $13.79/MMBtu. A cold-weather supply-and-demand shock is a plausible explanation: stronger heating demand, production disruptions, and pipeline constraints can jointly tighten markets. However, the uploaded Python file does not independently verify the storm name, weather severity, production outages, or the causal chain. Those claims should be checked against contemporaneous EIA and weather reports before publication.

Limitations and next steps

East Region storage is not a nationwide storage measure, while Henry Hub is a benchmark located in Louisiana. Weekly averages can conceal extreme daily moves; the January spike should therefore be analyzed at both daily and weekly resolution. A stronger model would include heating degree days, Lower 48 storage, production, LNG feedgas, pipeline flows, and lagged variables, with an out-of-sample test.

Reproducibility

The accompanying cleaned Python script removes the exploratory commands and hard-coded API credentials, downloads data using an EIA API key entered securely, saves CSV outputs and charts, and recomputes correlation and regression results. Values above are transcribed from the user’s Spyder screenshots, not independently recalculated from fresh EIA responses in this environment.

Data source: U.S. Energy Information Administration (EIA), API v2. Analysis date: October 2026.